

Chethan S

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Profile

4th-year Computer Science Engineering student at PES University specializing in machine learning, uncertainty-aware AI, and scalable AI systems. Experienced in developing end-to-end ML pipelines, LLM-powered applications, and full stack systems, seeking AI/ML and Software Engineering roles.

Education

PES University , Bengaluru	2023 – Present
B.Tech in Computer Science Engineering	CGPA: 8.66 / 10
Gnana Dhama PU College	2021 – 2023
PCMB	94.5%
Vidya Spoorti International Academy	2016 – 2021
State Board	95.36%

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C

Frameworks & Libraries: PyTorch, LangGraph, scikit-learn, XGBoost, LightGBM, SHAP, Optuna, FAISS, sentence-transformers, React.js, Next.js, FastAPI, Node.js, Express.js

Databases: MongoDB, MySQL

Core Areas: Machine Learning, Deep Learning, Few-Shot Learning, Agentic RAG, LLM Evaluation, AI Systems, Full Stack Development

Tools & Technologies: Git, GitHub Actions (CI/CD), Linux, REST APIs, Ollama, Model Context Protocol (MCP), Plotly, Uvicorn

Projects

Uncertainty-Aware Few-Shot Learning with Model Agnostic Meta Learning 2026

Python | PyTorch | MAML | MC Dropout | ERCICAM-2026

- Developed an uncertainty-aware extension of MAML using MC Dropout and predictive entropy without modifying the underlying MAML architecture or optimization procedure
- Designed three parameter-free task-level diagnostics – Gradient Instability, Parameter Shift Magnitude, and Predictive Variance – to analyze reliability of few-shot adaptation
- Evaluated 3 CNN architectures across Omniglot and Fashion-MNIST under a 5-way 1-shot protocol using accuracy, ECE, NLL, Brier Score, and AUROC; presented the work at ERCICAM-2026

TraceRAG – Evaluation-Driven Agentic RAG Platform 2026

Python | LangGraph | FAISS | Ollama | FastAPI | React

- Built a local-first agentic RAG system as a LangGraph state machine with query analysis, evidence grading, adaptive query rewriting, and citation-grounded generation, running on Ollama (Qwen 2.5 / LLaMA 3.2) at zero API cost
- Implemented hybrid retrieval fusing FAISS dense embeddings with BM25 via Reciprocal Rank Fusion; exposed through FastAPI with rate limiting and API-key auth, an MCP server, and a React dashboard with reasoning-trace inspection
- Designed a 40-question LLM-as-a-Judge benchmark across 5 research domains against a baseline RAG: retrieval relevance improved from 50.0% to 93.1% and answer accuracy from 79.2% to 83.9% with 0% hallucination, at ~2x latency

AutoML Platform for Automated Model Selection & Hyperparameter Optimization 2026

Next.js | FastAPI | ML

- Developed a modular AutoML platform with a Next.js (TypeScript) dashboard and FastAPI backend supporting datasets up to 50 MB (~10k rows)
- Automated model selection across 16 classifiers and regressors with Optuna-driven hyperparameter tuning
- Integrated SHAP for model explainability and Plotly for interactive leaderboard visualizations; surfaced top-5 model recommendations by default
- Designed async training pipelines with persistent run artifacts (model, pipeline, and training report) stored per run

Leadership & Experience

Head of Operations – Equinox Club, PES University 2025 – 2026

- Led end-to-end execution of 4 technical events – including a 150-participant datathon and two engineering challenges – managing venue, sponsorship, and volunteer coordination
- Built and directed cross-functional teams of 10+ members across planning, logistics, and on-ground operations, delivering all events on schedule

Achievements & Certifications

- CNR Rao Scholarship** – awarded by PES University for academic merit (top percentile, B.Tech CSE cohort)
- Introduction to AppDev using Flutter & Dart** – PESU I/O